

Yash Rawal

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Professional Summary

Applied AI & Machine Learning Engineer serving a **1M+ user scale** by architecting enterprise agentic platforms, hierarchical RAG architectures, and automated intelligence pipelines across 120+ organizations. Specialized in Python, high-throughput LLM serving internals (vLLM contributor), workflow orchestration (SAP Joule 2.0, LangGraph), and automated testing/evaluation control planes. Bridges generative AI models and fine-tuning with resilient cloud-native infrastructure (Azure, AWS, Docker, Kubernetes).

Skills

Languages & Backend: Python, SQL, FastAPI, PostgreSQL, Oracle, REST APIs, WebSockets, AsyncIO, Linux

AI & ML Engineering: Agentic AI, Model Context Protocol (MCP), Multi-Agent Systems, Tool RAG, Healthcare NLP, vLLM Serving

AI & Workflow Frameworks: LangChain, LangGraph, vLLM, PyTorch, Hugging Face, SAP Joule 2.0, vLLM Tool Parsers, LlamaIndex, GROBID

Cloud & Infrastructure: Azure, AWS, Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins, Git, Gerrit), Redis, PGVector

Testing, QA & Security: PyTest, LLM-as-a-Judge, Prompt Lab, Red Teaming, PII Sanitization, Unit/Integration/Regression Testing

Work Experience

Bigscal Technologies Pvt. Ltd.

AI/ML Engineer

Oct 2024 - Present

Surat, India

- Deployed a production-scale agentic platform and multi-agent systems serving **1M+ active ERP users** across 120+ enterprise tenants on cloud infrastructure (Azure, AWS).
- Architected multi-agent routing with **LangGraph & SAP Joule 2.0**, implementing hybrid semantic-lexical tool discovery (vector + BM25F/RRF) across 650+ dynamic read/write and generation MCP capabilities.
- Designed a capability contract layer to execute nested profiles inside a single ReAct loop, reducing execution overhead by **3x** and token consumption by 70%.
- Trained domain-specific **E5-large-v2** embedding models on Lightcast labor-market datasets using PyTorch triplet contrastive learning to optimize semantic job-skill intent mapping.
- Implemented a 4-level hierarchical RAG pipeline in Python indexing 10,000+ technical documents using GROBID, pdfplumber, and PostgreSQL/PGVector, achieving a **90%+ Recall@10** retrieval rate.
- Developed the Prompt Lab evaluation control plane in Python to collect reasoning trajectories and human preference annotations, establishing a scalable **Human Feedback (RLHF) Pipeline**.

Bigscal Technologies Pvt. Ltd.

AI/ML Engineer Intern

Apr 2024 - Sep 2024

Surat, India

- Engineered clinical summarization pipelines in Python for 116K+ medical records with 4-bit quantized Llama 3, and designed real-time WebSocket audio streaming for Indic ASR with **NeMo RNN-T and CTC decoding**.
- Researched audio digital signal processing (STFT, Mel-spectrograms, MFCC feature extraction via librosa/torchaudio) and CNN audio architectures (VGGish, ResNet), compiling custom Indic LJSpeech-format datasets.

Projects

vLLM High-Performance Inference Engine (vllm-project/vllm) | Open Source

Merged **PR #55475** fixing structural XML tool parser silent drops on `v1/chat/completions` and Pydantic param indexing (64/64 Qwen tests and 983+ regression suite passed); engineered `torchcodec` multimodal decoder guards (Issue #54097).

Python, PyTorch, Pydantic, AsyncIO, PyTest

Job Framework Semantic Embedding Model (e5-finetuned-job-framework) | Model Training

Fine-tuned an ~335M-parameter **E5-large-v2** transformer on Lightcast labor-market data using triplet contrastive learning with hard-negative mining in PyTorch, replacing general-purpose embeddings to optimize semantic job-to-skills retrieval precision.

Python, PyTorch, Hugging Face, E5-large-v2

Modular Agent Harness | Project

Principal Creator & Architect of a modular, plug-and-play Python agentic framework implementing a formal ReAct loop with step budgets, pgvector tool routing, and dynamic MCP settings caching.

Python, FastAPI, PostgreSQL, WebSockets, pgvector

Education

Master of Computer Applications (Artificial Intelligence & Machine Learning) - Parul University

2022 - 2024

Certifications

Microsoft Certified: Azure AI Fundamentals (AI-900)

Microsoft

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Microsoft